

Evaluation Study

Ablation study across six retrieval configurations

Model: gemma4:e2b | 10 ground-truth queries | Corpus: 176 dense vectors / 77 BM25 documents

Metric	Dense Only	BM25 Only	Hybrid Static	Hybrid Adaptive	Hybrid + Reranker	Full Pipeline
RETRIEVAL METRICS						
Embedding time ↓	12.67 s	0.00 s	11.63 s	12.32 s	12.47 s	12.31 s
Retrieval latency ↓	0.063 s	0.004 s	0.048 s	0.052 s	0.169 s	0.171 s
Relevance – Text ↑	0.7503	0.6031	0.7471	0.7458	0.7423	0.7431
Relevance – Image ↑	0.4110	0.5773	0.4108	0.4043	0.4120	0.4042
Page Coverage@k ↑	90.00%	90.00%	92.50%	95.00%	90.00%	90.00%
GENERATION QUALITY						
ROUGE-1 ↑	0.5153	0.5074	0.5120	0.5138	0.5229	0.5124
ROUGE-2 ↑	0.1976	0.1904	0.1998	0.1987	0.2123	0.2022
ROUGE-L ↑	0.2878	0.2827	0.2938	0.2915	0.3056	0.2875
BLEU ↑	0.1307	0.1257	0.1274	0.1329	0.1458	0.1352
METEOR ↑	0.3353	0.3334	0.3304	0.3392	0.3466	0.3489
BERTScore F1 ↑	0.8763	0.8748	0.8777	0.8770	0.8793	0.8772
FCD ↓	17.87	18.56	18.29	16.96	19.09	18.89
Faithfulness ↑	87.86%	89.83%	95.83%	92.20%	93.57%	92.95%
GT Coverage ↑	89.77%	90.10%	91.90%	92.02%	92.18%	92.07%
PERFORMANCE						
End-to-end latency ↓	16.43 s	8.37 s	16.05 s	17.35 s	17.32 s	16.74 s
Throughput ↑	0.061 q/s	0.120 q/s	0.062 q/s	0.058 q/s	0.058 q/s	0.060 q/s
RAM usage ↓	0.125 GB	0.930 GB	0.125 GB	0.111 GB	0.126 GB	0.231 GB
GPU utilisation ↓	27.48%	41.12%	27.09%	26.49%	26.97%	27.32%

Configurations

Dense Only — semantic vector search only.
BM25 Only — lexical BM25 search only.
Hybrid Static — BM25 + semantic with fixed 0.5 / 0.5 weights.
Hybrid Adaptive — query-aware weight blending.
Hybrid + Reranker — static hybrid with cross-encoder reranking.
Full Pipeline — adaptive hybrid + reranker + distance filtering.

Metric Notes

↓ lower is better | ↑ higher is better.
FCD — Faithfulness Cosine Distance; divergence between response and prompt embeddings.
GT Coverage — fraction of ground-truth answer tokens present in retrieved context.
Retrieval metrics are model-independent and computed once per configuration before any LLM call.
Green cells indicate the best (optimal) value for that metric.